



Climatic Niche Divergence and Conservatism Promote Speciation in Snake-Eyed Skinks (Sauria: Scincidae): New Insight into the Evolution and Diversification of *Ablepharus* Species

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Received: 1 October 2022 / Accepted: 28 March 2023 / Published online: 7 April 2023

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Abstract

The degree to which closely related species conserve or diverge in their niche characteristics may provide insight into evolutionary and diversification processes, as well as some understanding of broad-scale biogeographic patterns. In this study, we used an ensemble of 10 algorithms to predict the distributions of 12 *Ablepharus* species in geographical (G)-space for the recent time and the future (2081–2100) climate conditions. Niche overlap, equivalency, and similarity tests in environmental (E)-space were used to test niche divergence and conservatism. The consensus model showed that temperature seasonality ($n=6$) and annual precipitation ($n=3$) were the major factors predicting the distribution of 75% (9 out of 12) of the species. The ensemble models based on ssp585 predict that 83.33% of the species ($n=10$) may have contracted their ranges, especially at lower altitudes along their southern margins. They may have also expanded northward during this period. However, in the ssp126, most species' distribution ranges may remain unchanged despite contractions and expansions. Based on multivariate niche analyses, the distributions of 28 and 11 paired comparisons explored in this study significantly diverged and conserved. We found that closely related species inhabit different habitats and respond to their climatic characteristics, indicating that ecology has a significant influence on speciation.

Keywords *Ablepharus* · Ecology · Distribution · Threats · Climate change · Conservation

Introduction

Ecological factors play a significant role in speciation by providing sources of selection that lead to microevolutionary change and by restricting organismal functions (Angert

& Schemske, 2005; Barrett & Hoekstra, 2011; Niewiarowski & Roosenburg, 1993). As a result, they have the power to influence the distributional limits of populations and species as well as phenotypic evolution and functional diversity (Sexton et al., 2009). Additionally, ecological variables may directly influence speciation (McKinnon et al., 2004). According to ecological speciation models, reproductive isolation emerges as an adaptive result of diverse natural selection (Rundle & Nosil, 2005; Schluter, 2001). In this context, speciation coexists with divergence in niche occupancy along at least one axis of multidimensional niche space (Cox et al., 2014; Gómez-Rodríguez et al., 2015). Even when non-ecological reasons produce reproductive isolation, multifarious selection can speed up and stabilize speciation, resulting in the secondary accumulation of ecological differences across lineages (Arnqvist et al., 2000; Burke & Arnold, 2001; Chevin et al., 2014; Mank, 2007; Owens et al., 1999). Furthermore, ecological restrictions may also contribute to speciation if evolutionary lineages separate geographically during a period of environmental

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change while retaining ancestral niche affinities (niche conservatism) (Wiens, 2004). In this instance, vicariant speciation occurs when ancestral distributional ranges split and shift (for example, during periods of orogeny or warming) as the new species cannot adapt to the novel environmental conditions that would otherwise enable gene flow to persist (Shepard & Burbink, 2008; Wiens, 2004). Therefore, functional lineage diversification and the level of niche similarity across members of a phylogenetic group should be influenced by the respective contributions of niche conservatism and niche divergence throughout speciation events (Culumber & Tobler, 2016). However, diversification still lacks a clear understanding of the relative importance of these components (Culumber & Tobler, 2016).

Ecologists and evolutionary biologists appreciate quantifying niche differentiation among closely related and parapatric species because it provides a solid basis for future experimental and observational work and raises questions about the “mechanistic underpinnings of broad-scale geographical patterns” (Wellenreuther et al., 2012). In the past, studies of the niche have required in-depth examinations of an organism’s needs for its local habitat (Galetto, 2005). Recently, regional environmental information has been made available due to the accessibility of global climate and land cover Geographic Information System (GIS) and remote-sensing data. Including these data in ecological niche modeling (ENMs hereafter) provides insight into ecological and evolutionary concerns by comparing niche similarities among species and predicting how they will respond to environmental changes (Broennimann et al., 2012; Guisan et al., 2014; Kozak et al., 2008; Soberón, 2007). There are, however, significant conceptual and statistical challenges associated with determining niche similarity based on occurrence records that cover a geographically representative range of species. For instance, most ENMs have environmental variables (such as temperature) that are frequently spatially correlated, so significant niche divergence may be confused with geographical distance (McCormack et al., 2010). The spatial autocorrelation problem cannot be ignored when comparing niche conservatism versus divergence, however, it may be addressed by using null models (McCormack et al., 2010; Warren et al., 2008). Realized environmental niches may be directly measured and compared in the environmental (E)-space using multivariate statistics to satisfy the growing demand for robust approaches for understanding niche differences in evolutionary and community contexts (Broennimann et al., 2012). Recent ordination null tests can evaluate niche conservatism or divergence hypotheses by applying a kernel estimation to smooth out biases caused by inevitable sampling efforts (Broennimann et al., 2012).

The small, secretive, and oviparous snake-eyed skinks in the genus *Ablepharus* Fitzinger, 1823 (Family: Scincidae) are an excellent study system for investigating the impacts of climate on genetic structure as well as how divergence and subsequent speciation processes have been driven over time. They show a wide geographical distribution in southeastern Europe, southwest Asia, and central Asia (Karamiani et al., 2022; Mirza et al., 2022) and occur in different environmental conditions including sub-Mediterranean deciduous forests, shrubs, and steppes with limestone, sandstone, basalt, loess, and calcareous sand substrates (Herczeg et al., 2004; Jovanovic Glavas et al., 2022; Karamiani et al., 2022; Ljubisavljević et al., 2002). This genus contains 11 species (Karamiani et al., 2022), however, a recent systematic study found it to be synonymous with *Asymblepharus* (Mirza et al., 2022). This resulted in the transfer of eight species from *Asymblepharus* to *Ablepharus* (Mirza et al., 2022). These species’ habitat requirements and potential distributions under climate change, as well as the role of climate influences in their speciation, have received comparatively little attention. In this sense, Sanchooli (2016) investigated the distribution of *A. bivittatus* in Iran and concluded that *A. bivittatus* prefers suitable habitats in the northern regions due to habitat competition with *A. pannonicus*. Karamiani et al. (2018) explored the potential distribution of *A. grayanus* and *A. pannonicus* species in the present as well as the mid-Holocene and Last Glacial Maximum periods. These authors concluded that these two species had a wider and more suitable distribution range in the past compared to the present time. Kurnaz and Yousefkhani (2021) also evaluated the ecological niches of *A. budaki* and *A. anatolicus* in terms of biogeography and found that both species had different niche requirements. Finally, Vaissi (2022a) projected *A. pannonicus* distribution in Iran in the present and the future. As a result of this study, *A. pannonicus* will probably lose habitat on the eastern and southern margins, and its distribution patterns may shift to higher elevations.

The study of niche relationships can provide new insights into ecological distinctiveness and the mechanisms behind species evolution (McCormack et al., 2010; Raxworthy et al., 2007; Schluter, 2009; Wiens & Graham, 2005). Our study explores the relationship between climatic niche variation and speciation. This addresses the question of whether *Ablepharus* species maintained the same niche, as well as whether niche divergence or niche conservation influenced their diversification process. We also investigate the relationships between recent *Ablepharus* distributions and observed climate, as well as potential future species distributions. For this purpose, ENMs based on climatic variables were developed in concert with occurrence data in the geographic space (G-space hereafter) to project species’ recent and future distribution range (2081–2100). We then used

an ordination null test of PCA-environment in E-space to assess whether these closely related species occupy habitats that are more similar or more different than expected based on background environmental divergence. These results provide a multidimensional and comprehensive picture of niche variation within *Ablepharus* species when combined with phylogeographic relationships (Bozkurt & Olgun, 2020; Karamiani et al., 2021; Mirza et al., 2022; Poulakakis et al., 2005, 2013; Skourtanioti et al., 2016). Furthermore, recent conservation efforts and future research into the impacts of climate change on biodiversity depend on a knowledge of the primary ecological restrictions on species distribution (Aguirre-Gutiérrez et al., 2015).

Materials and methods

Study area and species data

Figure 1 displays the study area, which included the entire range of *Ablepharus* species, and covers Asia and southern Europe, as well as possibly northern Africa (Anderson,

1999; Fuhn, 1969; Karamiani et al., 2018; Khan, 2002; Vyas, 2011). We used 12 of the 19 known species in this study because the occurrence records were sufficient for the analysis (Table 1). We obtained occurrence records from a variety of sources; detailed information is available in Table S1. Then, we removed records that were duplicates, had dubious taxonomic identification, or had ambiguous localities (such as the centers of countries or states). Finally, we used occurrence data that were more than 500 m apart in order to reduce the impacts of spatial autocorrelation (Boria et al., 2014) using the R package spThin (Aiello-Lammens et al., 2015). Table 1 lists the species, their conservation status, and the number of occurrence records before and after the analysis. Besides Table S1, Fig. 1 also displays the geographic coordinates of these occurrences.

Climatic variables

Six climatic variables were employed to predict the species niche under recent (1970–2000) and future (2081–2100) climate change projections: annual mean temperature (BIO1 hereafter); temperature seasonality (BIO4 hereafter); the

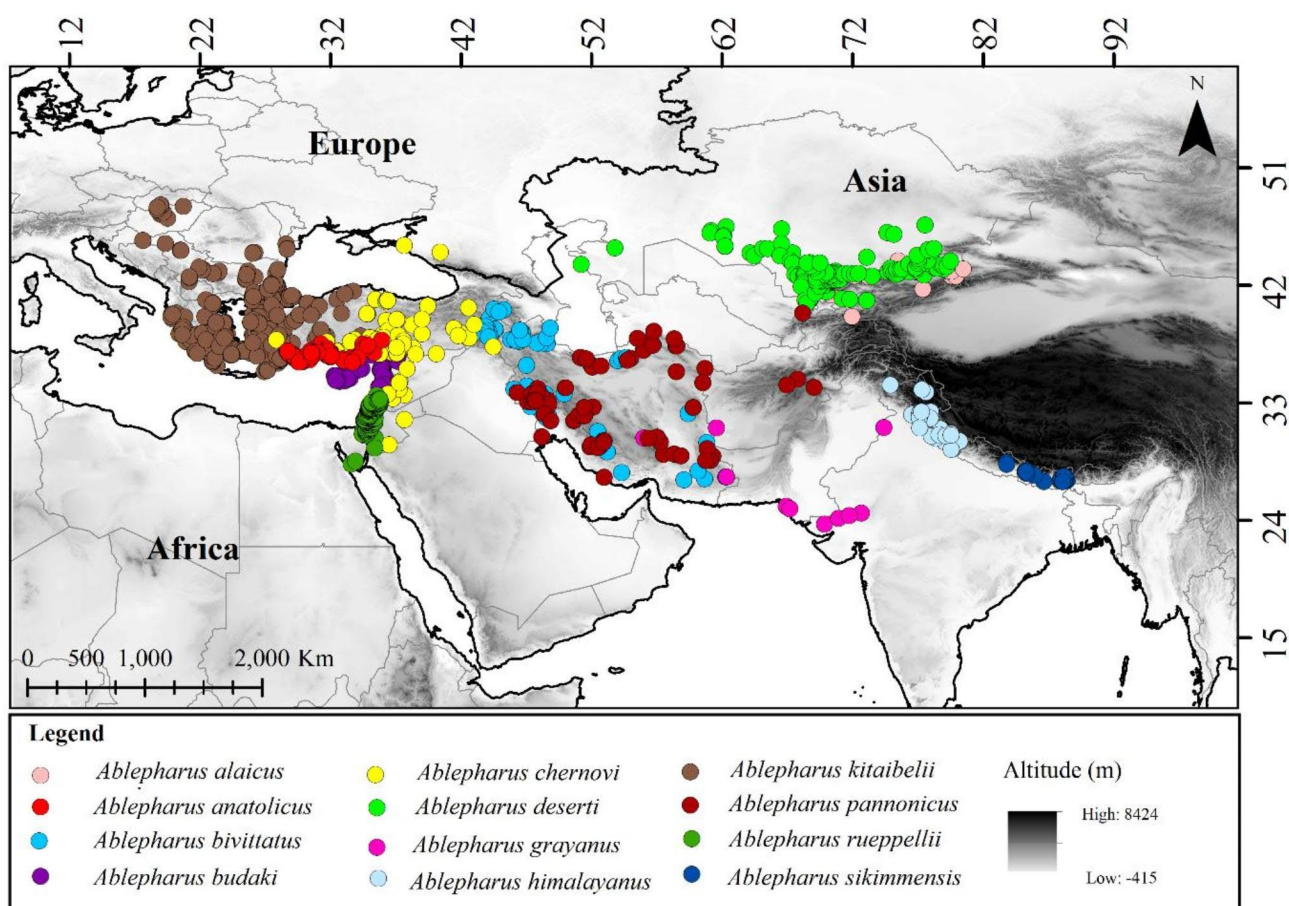


Fig.1 Study area. The occurrence records of 12 species of *Ablepharus* with different colours are shown on the map

Table 1 The number of *Ablepharus* species (N = 12) occurrence records before and after correcting to a resolution of 30 arc seconds (1 km), conservation status under IUCN criteria as well as the quality of ensemble models. TSS: true skill statistic, AUC: area under the receiver operating characteristic curve, and KAPPA: Cohen's Kappa. LC: Least Concern. NA: Not Evaluated

No.	Species	IUCN status	Occurrence records (N)		Ensemble models quality		
			Before correcting	After correcting	TSS	AUC	KAPPA
1	<i>Ablepharus alaicus</i>	LC	50	35	0.98	1	0.95
2	<i>Ablepharus anaticus</i>	NA	74	69	0.99	1	0.96
3	<i>Ablepharus bivittatus</i>	LC	121	116	0.90	0.98	0.82
4	<i>Ablepharus budaki</i>	LC	100	91	0.98	1	0.97
5	<i>Ablepharus chernovi</i>	LC	75	74	0.94	0.99	0.88
6	<i>Ablepharus deserti</i>	LC	126	117	0.92	0.99	0.88
7	<i>Ablepharus grayanus</i>	LC	12	12	0.93	0.99	0.87
8	<i>Ablepharus himalayanus</i>	LC	31	28	0.96	1	0.96
9	<i>Ablepharus kitaibelii</i>	LC	457	388	0.97	1	0.96
10	<i>Ablepharus pannonicus</i>	LC	71	69	0.93	0.99	0.83
11	<i>Ablepharus rueppellii</i>	LC	247	204	0.99	1	0.99
12	<i>Ablepharus sikimensis</i>	LC	25	24	0.99	1	0.98

max temperature of the warmest month (BIO5 hereafter), annual precipitation (BIO12 hereafter); precipitation of driest month (BIO14 hereafter); and precipitation seasonality (BIO15 hereafter). These bioclimatic variables were downloaded from the WorldClim v2.1 (<https://www.worldclim.org>) with 30-second spatial resolution raster grids. BIO1 and BIO12 were selected as they were found to have the most significant effects on reptile richness and range distribution in the study region (Kafash et al., 2020; Vaissi, 2022). BIO4, BIO5, BIO14, and BIO15 were selected because they are biologically significant, have weak global correlations, and may reveal environmental variables that impact distributions (Jarvie & Svenning, 2018).

The future data were obtained by averaging two global climate models (GCMs) from the Goddard Institute for Space Studies and Meteorological Research Institute for the years 2081–2100: GISS-E2-1-G (Kelley et al., 2020) and MRI-ESM2-0 (Yukimoto et al., 2019). These models have the best performance among the 32 models for Eurasia (Sun et al., 2022). The lowest and the highest limits of the shared socioeconomic pathways (ssps) from the Coupled Model Intercomparison Project Phase 6 (CMIP6) (Eyring et al., 2016), 126 (lowest emissions, smallest temperature increase) and 585 (highest emissions, largest temperature increase), were analyzed to predict future climate conditions.

Species distribution modelling

We used the “biomod2” package (v 3.4.6) in the R (v 4.2.0) programming language and followed default settings recommended by Guisan et al. (2018) in the G-space to simulate species distribution using an ensemble of 10 algorithms (Thuiller et al., 2016). These algorithms were Artificial Neural Networks (ANN, CV.ann = 5, 5 Fold Cross

Validation + 3 Repetitions), Classification Tree Analysis (CTA, CV.tree = 50, 5 Fold Cross-Validation), Flexible Discriminant Analysis (FDA), Generalized Additive Models (GAM, algo = 'GAM_mgcv'), Generalized Boosted Models (GBM, n.trees = 1000, 3 Fold Cross-Validation), Generalized Linear Models (GLM, type = 'quadratic', interaction.level = 1, the stepwise procedure using Akaike Information Criterion (AIC) criteria), Maximum Entropy (MaxEnt, Phillips, maximum iterations = 500, https://biodiversityinformatics.amnh.org/open_source/maxent/), Multivariate Adaptive Regression splines (MARS, simple with no interaction), Random Forest (RF, n.trees = 1000), and Surface Range Envelops (SRE, quant = 0.025) (Guisan et al., 2018).

These modeling algorithms need both presence and absence datasets, but it might be difficult to find the real absence data. Thus, following Wisz and Guisan (2009), Barbet-Massin et al. (2012), and Guisan et al. (2018), we generated 500 pseudo-absences at random within the study area. We repeated the procedure three times to address the potential sample bias in the pseudo-absence generation because this process of pseudo-absence generation is a stochastic procedure caused by the random selection of the pseudo-absences (Allouche et al., 2006). The models are calibrated on 80% of the data (training set) and evaluated on the remaining 20% (validation set). The entire procedure is repeated four times. The mean of the important variables was determined for the models. The higher mean indicates the variable is more significant (Guisan et al., 2018). By default, the True Skill Statistics (TSS), Area Under Curve-Receiver Operating Characteristics statistics (AUC), and Cohen's kappa (KAPPA) metrics are used to evaluate each model (Allouche et al., 2006; Cohen, 1960; Fielding & Bell, 1997; Hanley & McNeil, 1982). The TSS and KAPPA values range from −1 to +1, with 1 indicating perfect agreement and 0.60 to 0.90 indicating fair to good

model performance (Allouche et al., 2006). AUC values of 0.90 and above are regarded as good, 0.60 and 0.90 are considered medium, while 0.60 and below are considered poor (Phillips et al., 2006).

Ensemble models are created using the BIOMOD EnsembleModeling() function. We consider two “ensembling” options—committee averaging and weighted mean—to reduce the number of outputs (Guisan et al., 2018). We mix all models (i.e. all techniques, all pseudo-absences sampling, and all cross-validation runs) to produce our ensembles of models (Guisan et al., 2018). TSS was used as the evaluation reference for committee building and defining weights. This means that only models with a TSS greater than or equal to 0.8 are kept to build the final ensemble (Guisan et al., 2018).

Species range change

We evaluated and displayed the range changes over future climate scenarios for each of the species using the BIOMOD (Range Size) function in the biomod2 package. This function generates two outputs: a spatial map that highlights where species may gain or lose suitable conditions, and a table with summary data of species range change. More specifically, from both output types, we can get information about four absolute metrics: “Loss” denotes the number of pixels that the studied species is predicted to lose as a result of climate change; “Absent” denotes the number of pixels that the studied species is currently not occupying and that are also predicted to be unsuitable under a specific climatic scenario; and “Stable” denotes the number of pixels that the studied species is currently occupying and that are also predicted to remain occupied into the future, and “Gain,” which indicates the number of pixels that the studied species is currently not occupying but are expected to be occupied in the future. Finally, from these four metrics, three additional relative metrics were obtained, including “Percentage loss,” which corresponds to the percentage of currently occupied sites to be lost and is calculated as $(\text{Loss}/(\text{Loss} + \text{Stable}))$; “Percentage gain,” which corresponds to the percentage of new sites taking into account the species’ current distribution size and calculated as $(\text{Gain}/(\text{Loss} + \text{Stable}))$, and “Range change,” which represents the overall projection outcome and is equal to percentage gain – percentage loss (Guisan et al., 2018).

Niche analyses

Niche analysis was carried out separately for each of the 12 *Ablepharus* species using the R package ecospat (Di Cola et al., 2017). The E-space was evaluated utilizing a PCA-environment (PCA-env hereafter) framework and an ordination

approach. Climatic variables are changed into two-dimensional spaces defined by two principal components. The two-dimensional environmental space is then projected onto grid cells with a diameter of 100×100 and bounded by minimum and maximum PCA values in the background. A kernel density function is used to estimate a smoothed density of occurrences for each species in each grid cell. The overlap between niches was calculated using Schoener’s D and Hellinger’s I metrics, which range from 0 (no overlap) to 1 (complete overlap) (Broennimann et al., 2012).

We performed equivalency and similarity tests to compare the overlap of the two observed niches to the overlap of simulated niches to test if niche overlap was statistically significant (Schoener’s D). The niche equivalency test determines if two niches are equivalent by randomly permuting occurrences between ranges. The niche similarity test determines if the species’ niches are more or less similar than anticipated by chance, using random shifts of niches within accessible conditions in the study region (Broennimann et al., 2012; Di Cola et al., 2017; Warren et al., 2008). We tested niche conservatism (alternative = higher, i.e. the niche overlap is more equivalent/similar than random) and niche divergence (alternative = lower, i.e. the niche overlap is less equivalent/similar than random) by the “alternative” option at ecospat (Broennimann et al., 2012; Di Cola et al., 2017). To test each hypothesis, we performed 1000 permutations. R (v 4.2.0) was used to perform all of the analyses (www.r-project.org).

Results

Species distribution modelling

TSS, AUC, and KAPPA values ranging from 0.82 to 1 indicate high-quality ensemble models (Table 1). Figure 2 shows the map of suitable habitats for 12 *Ablepharus* species over their entire range, based on six climatic factors for recent and future climate projections (2081–2100) under SSP126 and SSP585 scenarios. For the 12 *Ablepharus* species, Table 2 provides the mean of variable importance (%) predicted by the 10 algorithms. The most important variables impacting the distribution of species were determined to be BIO4 for six species, BIO12 for three species, BIO5 for two species, and BIO15 for one species (Table 2). Climate change will have a range of impacts on *Ablepharus* species, with some increasing their ranges, some decreasing their ranges, and others staying relatively unaffected (Fig. 2, Fig. S1, Table 3). Changes in the ranges of *Ablepharus* species over time are shown in Fig. S1 and Table 3, including habitat stability, habitat gain or expansion, and habitat loss

Fig. 2 Recent and future (2081–2100) habitat suitability for 12 species of *Ablepharus* based on the consensus model under optimistic (ssp126) and pessimistic (ssp585) scenarios. **A:** *Ablepharus alaicus*; **B:** *Ablepharus anatolicus*; **C:** *Ablepharus bivittatus*; **D:** *Ablepharus budaki*; **E:** *Ablepharus chernovi*; **F:** *Ablepharus deserti*; **G:** *Ablepharus grayanus*; **H:** *Ablepharus himalayanus*; **I:** *Ablepharus kitaibelii*; **J:** *Ablepharus pannonicus*; **K:** *Ablepharus rueppellii*; **L:** *Ablepharus sikimensis*

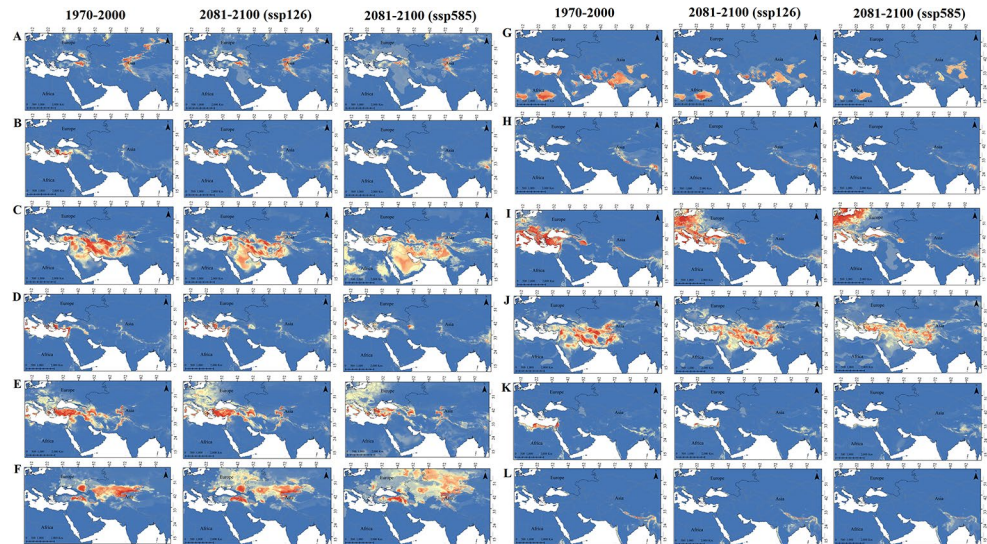


Table 2 Mean of variable importance (%) by the 10 algorithms for the 12 species of *Ablepharus*

No.	Species	BIO1	BIO4	BIO5	BIO12	BIO14	BIO15
1	<i>Ablepharus alaicus</i>	20.37	26.97	14.72	5.45	19.49	12.96
2	<i>Ablepharus anatolicus</i>	20.11	19.83	20.73	20.21	8.97	10.12
3	<i>Ablepharus bivittatus</i>	14.48	37.72	12.48	10.71	8.15	16.04
4	<i>Ablepharus budaki</i>	22.01	19.69	13.71	25.20	11.79	7.57
5	<i>Ablepharus chernovi</i>	17.44	24.84	18.48	14.88	11.82	12.50
6	<i>Ablepharus deserti</i>	23.76	24.29	14.83	8.66	10.05	18.39
7	<i>Ablepharus grayanus</i>	22.60	18.81	15.74	7.16	9.47	26.20
8	<i>Ablepharus himalayanus</i>	15.32	15.13	14.50	30.42	15.51	9.08
9	<i>Ablepharus kitaibelii</i>	21.47	31.36	13.85	21.53	5.02	6.73
10	<i>Ablepharus pannonicus</i>	20.59	30.40	14.72	13.18	7.68	13.40
11	<i>Ablepharus rueppellii</i>	24.00	14.68	28.00	2.56	14.90	15.83
12	<i>Ablepharus sikimensis</i>	17.74	11.37	18.92	32.65	46.60	14.70

Annual mean temperature (BIO1); temperature seasonality (BIO4); max temperature of warmest month (BIO5); annual precipitation (BIO12); precipitation of driest month (BIO14); and precipitation seasonality (BIO15)

Table 3 Species range change (SRC) of 12 species of *Ablepharus* in recently suitable habitats (gain/loss) by 2081–2100 under optimistic (ssp126) and pessimistic (ssp585) scenarios

No.	Species	ssp126			ssp585		
		Lost	Gain	SRC	Lost	Gain	SRC
1	<i>Ablepharus alaicus</i>	33.34	17.89	-15.45	52.19	26.94	-25.24
2	<i>Ablepharus anatolicus</i>	35.94	18.26	-17.67	91.96	35.36	-56.60
3	<i>Ablepharus bivittatus</i>	17.97	30.94	12.96	26.55	91.59	65.03
4	<i>Ablepharus budaki</i>	30.47	30.16	-0.30	83.12	37.68	-45.44
5	<i>Ablepharus chernovi</i>	30.35	22.49	-7.46	62.62	23.49	-39.13
6	<i>Ablepharus deserti</i>	6.89	74.00	67.11	14.97	177.81	162.83
7	<i>Ablepharus grayanus</i>	26.26	4.86	-21.40	65.47	14.82	-50.65
8	<i>Ablepharus himalayanus</i>	49.05	4.09	-44.95	75.53	3.94	-71.89
9	<i>Ablepharus kitaibelii</i>	22.78	67.66	44.87	72.21	91.40	19.19
10	<i>Ablepharus pannonicus</i>	31.17	23.16	-8.01	54.75	34.76	-19.99
11	<i>Ablepharus rueppellii</i>	55.39	9.51	-45.87	83.45	15.37	-68.07
12	<i>Ablepharus sikimensis</i>	39.49	10.02	-29.47	76.86	6.42	-70.44

or contraction. The following are the details responses of 12 species to climate change:

A. alaicus BIO4 (26.97%) and BIO1 (20.37%) are the two most important variables affecting the *A. alaicus* distribution, respectively (Table 2). The habitat suitability map under recent climate conditions indicates that the eastern edges to the southeast of Kazakhstan, Kyrgyzstan, Tajikistan (except southwest), northeast of Pakistan, and the northern margin of the Kunlun and Qilian Mountains in China have medium to high habitat suitability (Fig. 2A). According to the models, eastern Türkiye, Armenia, and small parts of southwest and southeast Russia also showed high habitat suitability (Fig. 2A). In the future, the distribution range of *A. alaicus* may contract at the margins of its range and may expand toward the north, but these changes may be relatively small, with most of the existing range remaining unchanged (Table 3 and Fig. S1A).

A. anatolicus BIO5 (20.73%) and BIO12 (20.21%) are the two most important variables affecting the *A. anatolicus* distribution, respectively (Table 2). The habitat suitability map under recent climate conditions indicates that countries around the Mediterranean, including southern Greece, Cyprus, the west, southwest, and southern Türkiye, western Syria, and Lebanon have high habitat suitability (Fig. 2B). Climate change would not significantly affect the *A. anatolicus* distribution under the ssp126, however, the ssp585 may result in the loss of 91.96% of the species' suitable habitats in the distribution region (Table 3 and Fig. S1B).

A. bivittatus BIO4 (37.72%) and BIO15 (16.04%) are the two most important variables affecting the *A. bivittatus* distribution, respectively (Table 2). The habitat suitability map under recent climate conditions indicates that Iran (except central parts), Afghanistan (except southwest), southeastern Turkmenistan to southwestern Uzbekistan, west Tajikistan, Azerbaijan, Armenia, Türkiye (except northern strip), Syria, western and eastern Iraq to northern Saudi Arabia, and Kuwait have high habitat suitability (Fig. 2C). ssp126 predicts that the species' distribution may remain stable; however, ssp585 predicts 91.59% of the species habitats may be gained at high and especially low latitudes, such as north Africa (Table 3 and Fig. S1C).

A. budaki BIO12 (25.20%) and BIO1 (22.01%) are the two most important variables affecting the *A. budaki* distribution, respectively (Table 2). The habitat suitability map under recent climate conditions indicates that the countries around the Mediterranean, including southern Greece, Cyprus, the west, southwest, and southern margins of Türkiye, western Syria, Lebanon, and northern Palestine/

Israel, as well as a portion of South Uzbekistan, Tajikistan, eastern Afghanistan, and northern Pakistan have medium to high habitat suitability (Fig. 2D). Climate change causes habitat gain at high latitudes in both scenarios, though most *A. budaki* habitats remain stable under the ssp126. In the ssp585, habitat losses (83.12%) may be greatest in the countries surrounding the eastern Mediterranean as well as southern Türkiye (Table 3 and Fig. S1D).

A. chernovi BIO4 (24.84%) and BIO5 (18.48%) are the two most important variables affecting the *A. chernovi* distribution, respectively (Table 2). The habitat suitability map under recent climate conditions indicates that Northern and eastern Mediterranean countries, Türkiye, the Caucasus, Iran (except central and southern regions), northern and eastern Afghanistan, north and northwest of Pakistan to Tajikistan and southern Kazakhstan, and southeast Uzbekistan have high habitat suitability (Fig. 2E). The majority of the recent distribution range may be preserved by future climate under the ssp126, although the southern margin at lower latitudes may be lost especially under the ssp585 (62.62%). On the other hand, the northern margin at higher latitudes may gain new habitat (Table 3 and Fig. S1E).

A. deserti BIO4 (24.29%) and BIO1 (23.76%) are the two most important variables affecting the *A. deserti* distribution, respectively (Table 2). The habitat suitability map under recent climate conditions indicates that all of Kazakhstan's southern areas, a small portion of southwestern Russia, western Turkmenistan, north and northwest of Uzbekistan, west Kyrgyzstan, eastern Türkiye, western Armenia, and northwest, parts of the north and northeast of Iran have high habitat suitability (Fig. 2F). The majority of the recent distribution range may be preserved due to the effects of future climate on both scenarios (especially ssp126), but southern margins at lower latitudes will be lost (especially ssp585: 14.97%) and northern margins at higher latitudes may gain new habitat (especially ssp585: 177.81%) (Table 3 and Fig. S1F).

A. grayanus BIO15 (26.20%) and BIO1 (22.60%) are the two most important variables affecting the *A. grayanus* distribution, respectively (Table 2). The habitat suitability map under recent climate conditions indicates that major north and northwest India, east Pakistan, a small part of the west and south Afghanistan, parts of the south and southeast of Iran, small parts of the central and southern regions of Saudi Arabia, east and south margins of the Mediterranean, western Sudan, eastern Chad, and southern Niger have high habitat suitability (Fig. 2G). The majority of this species distribution range may be lost at the southern margin,

especially under the ssp585 (65.47%) due to future climate change (Table 3 and Fig. S1G).

A. himalayanus BIO12 (30.42%) and BIO14 (15.51%) are the two most important variables affecting the *A. himalayanus* distribution, respectively (Table 2). The habitat suitability map under recent climate conditions indicates that east Tajikistan, a small part of eastern Afghanistan, northern borders of Pakistan and India, Nepal, northern borders of Bhutan and Myanmar have high habitat suitability (Fig. 2H). Most of the species' distribution range may be lost at the margins due to future climate change, especially under the ssp585 (75.53%) (Fig. S1H).

A. kitaibelii BIO4 (31.36%) and BIO12 (21.53%) are the two most important variables affecting the *A. kitaibelii* distribution, respectively (Table 2). The habitat suitability map under recent climate conditions indicates that northern and eastern Mediterranean countries, southern, western, and northern Black Sea border, southern European countries toward central Europe, northern Iran, Georgia, and northern and western borders of Azerbaijan have high habitat suitability (Fig. 2I). The majority of the recent distribution range may be preserved due to the effects of future climate under the ssp126, but southern margins at lower latitudes may be lost under the ssp585 (72.21%) and northern margins at higher latitudes may gain new habitat (especially ssp585: 91.40%) (Table 3 and Fig. S1F).

A. pannonicus BIO4 (30.40%) and BIO1 (20.59%) are the two most important variables affecting the *A. pannonicus* distribution, respectively (Table 2). The habitat suitability map under recent climate conditions indicates that Iran (except narrow strip in the southeast and central plains), Azerbaijan, southwest to southeast Turkmenistan, southwest Tajikistan, southeast Uzbekistan, northern and eastern Afghanistan, Pakistan (except southern areas), Iraq (except southern areas), Syria, northern Jordan, northern Saudi Arabia, Türkiye (except northern strip) have high habitat suitability (Fig. 2J). Although climate change may result in habitat loss at the distribution margins (especially ssp585: 54.75%) and habitat gains at high and low latitudes (especially ssp585: 34.76%), the majority of the distribution range of this species may be preserved (especially ssp126) (Table 3 and Fig. S1J).

A. rueppellii BIO5 (28.00%) and BIO1 (24.00%) are the two most important variables affecting the *A. rueppellii* distribution, respectively (Table 2). The habitat suitability map under recent climate conditions indicates that western Mediterranean countries including southern Lebanon, Israel/Palestine, and western Jordan as well as a strip of southern

Mediterranean coastal countries in North Africa (Egypt and Libya) have high habitat suitability (Fig. 2K). Future climate change may lose the distribution range of this species at the southern margins, especially under the ssp585 scenario (83.45%) (Table 3 and Fig. S1K).

A. sikimensis BIO12 (31.61%) and BIO5 (18.92%) are the two most important variables affecting the *A. sikimensis* distribution, respectively (Table 2). The habitat suitability map under recent climate conditions indicates that a small part of northern India, Nepal, Bhutan, and north and west Myanmar have high habitat suitability (Fig. 2L). Future climate change may lead to the loss of the majority of the distribution range of this species at the margins, especially under the ssp585 (76.86%) (Table 3 and Fig. S1L).

Niche analyses

The sum of the first two axes of the PCA-env explained between 69.55% and 92.61% variation in the environmental study for all pairwise comparisons among *Ablepharus* species (Table 4, Fig. S3). The D and I metric was used to analyze pairwise niche overlap, which ranged from no (D and I = 0) or very limited (D and I = 0.004–0.1), moderate (D and I = 0.11–0.51) to substantial overlap (D and I = 0.52–0.76) (Table 4, Fig. S2). Table 4 summarizes the niche equivalency and similarity tests. In the niche equivalency test, the hypothesis of niche divergence was accepted ($p < 0.05$) in 28 pairwise comparisons among *Ablepharus* species, whereas the hypothesis of niche conservatism was accepted in 11 paired comparisons (Table 4). In the niche similarity test, only the 21 paired comparisons for conservatism hypotheses showed significant values (Table 4).

Discussion

In this study, the climatic niche variations among *Ablepharus* species were explained using ENMs and multivariate niche analyses. We specifically explored the question of whether the *Ablepharus* species evolve by maintaining different ecological niches (divergence) or by inhabiting a similar ecological niche (conservatism). Based on the overall pattern of niche conservatism and niche divergence, as well as the distinct evidence that temperature and precipitation variables account for the crucial differences between the climate niches of the *Ablepharus* species, it is hypothesized that climate-mediated adaptation is a significant factor in the speciation process in these snake-eyed skinks.

Migration movements of reptiles are remarkably limited compared to other large vertebrate groups (Hickling et al.

Table 4 Schoener's D and Hellinger's I index and *p*-values for the equivalency (E) and similarity (S) test for niche divergence (D) and conservatism (C). The correlation circle illustrates the changing relevance between two species ranges along the first two principal axes

No.	Species	Schoener's D	Hellinger's I	E (D)	S (D)	E (C)	S (C)	Correlation circle (%)	
								PC1	PC2
1	<i>Ablepharus alaicus</i> vs. <i>A. anatolicus</i>	0.01	0.01	0.007	0.78	0.99	0.22	81.89	9.07
2	<i>Ablepharus alaicus</i> vs. <i>A. bivittatus</i>	0.17	0.25	0.001	0.77	1	0.19	74.87	9.99
3	<i>Ablepharus alaicus</i> vs. <i>A. budaki</i>	0	0	0.97	0.95	1	1	69.68	16.22
4	<i>Ablepharus alaicus</i> vs. <i>A. chernovi</i>	0.14	0.20	0.001	0.94	1	0.06	55.94	19.49
5	<i>Ablepharus alaicus</i> vs. <i>A. deserti</i>	0.21	0.37	0.001	0.86	1	0.13	49.49	30.97
6	<i>Ablepharus alaicus</i> vs. <i>A. grayanus</i>	0	0	1	1	0.14	0.01	71.49	16.47
7	<i>Ablepharus alaicus</i> vs. <i>A. himalayanus</i>	0.01	0.02	0.001	0.78	1	0.25	63.85	19.01
8	<i>Ablepharus alaicus</i> vs. <i>A. kitaibelii</i>	0.02	0.03	0.19	0.70	0.88	0.29	60.58	17.75
9	<i>Ablepharus alaicus</i> vs. <i>A. pannonicus</i>	0.01	0.01	0.005	0.77	0.99	0.23	81.89	9.07
10	<i>Ablepharus alaicus</i> vs. <i>A. rueppellii</i>	0	0	0.93	1	1	1	75.56	16.82
11	<i>Ablepharus alaicus</i> vs. <i>A. sikimensis</i>	0	0	0.93	1	1	1	70.08	20.67
12	<i>Ablepharus anatolicus</i> vs. <i>A. bivittatus</i>	0.04	0.06	0.006	0.61	1	0.37	56.55	31.83
13	<i>Ablepharus anatolicus</i> vs. <i>A. budaki</i>	0.71	0.76	1	1	0.001	0.001	51.03	19.8
14	<i>Ablepharus anatolicus</i> vs. <i>A. chernovi</i>	0.29	0.46	0.001	0.80	1	0.17	57.79	23.71
15	<i>Ablepharus anatolicus</i> vs. <i>A. deserti</i>	0.06	0.08	0.001	0.73	1	0.26	64.02	28.59
16	<i>Ablepharus anatolicus</i> vs. <i>A. grayanus</i>	0.0004	0.0006	0.05	0.76	0.91	0.23	62.52	23.44
17	<i>Ablepharus anatolicus</i> vs. <i>A. himalayanus</i>	0	0	0.93	1	1	1	55.71	34.2
18	<i>Ablepharus anatolicus</i> vs. <i>A. kitaibelii</i>	0.48	0.69	0.99	0.99	0.009	0.006	66.29	15.21
19	<i>Ablepharus anatolicus</i> vs. <i>A. pannonicus</i>	0.03	0.03	0.02	0.87	0.97	0.12	44.21	38.74
20	<i>Ablepharus anatolicus</i> vs. <i>A. rueppellii</i>	0.12	0.15	0.001	0.86	1	0.12	62.95	17.78
21	<i>Ablepharus anatolicus</i> vs. <i>A. sikimensis</i>	0.02	0.02	0.003	0.98	0.99	0.19	50.90	38.88
22	<i>Ablepharus bivittatus</i> vs. <i>A. budaki</i>	0.15	0.19	0.001	1	1	0.004	53	28.02
23	<i>Ablepharus bivittatus</i> vs. <i>A. chernovi</i>	0.56	0.60	1	0.97	0.001	0.03	58.78	18.53
24	<i>Ablepharus bivittatus</i> vs. <i>A. deserti</i>	0.52	0.57	1	1	0.001	0.002	61.61	20.05
25	<i>Ablepharus bivittatus</i> vs. <i>A. grayanus</i>	0.55	0.58	0.91	0.94	0.05	0.03	62.35	19.02
26	<i>Ablepharus bivittatus</i> vs. <i>A. himalayanus</i>	0.05	0.06	0.14	1	0.86	0.08	51.57	37.54
27	<i>Ablepharus bivittatus</i> vs. <i>A. kitaibelii</i>	0.14	0.15	0.02	1	0.98	0.003	51.43	30.26
28	<i>Ablepharus bivittatus</i> vs. <i>A. pannonicus</i>	0.66	0.81	0.55	1	0.42	0.001	60.41	15.67
29	<i>Ablepharus bivittatus</i> vs. <i>A. rueppellii</i>	0.11	0.20	0.001	1	1	0.003	56.11	28.25
30	<i>Ablepharus bivittatus</i> vs. <i>A. sikimensis</i>	0	0	1	1	0.009	0.34	47.11	42.17
31	<i>Ablepharus budaki</i> vs. <i>A. chernovi</i>	0.35	0.45	0.07	0.56	0.92	0.41	50.75	21.35
32	<i>Ablepharus budaki</i> vs. <i>A. deserti</i>	0.06	0.08	0.001	0.52	1	0.45	59.06	27.38
33	<i>Ablepharus budaki</i> vs. <i>A. grayanus</i>	0.11	0.12	0.36	0.77	0.70	0.22	51.20	22.81
34	<i>Ablepharus budaki</i> vs. <i>A. himalayanus</i>	0.01	0.02	0.001	0.68	1	0.34	45.35	33.26
35	<i>Ablepharus budaki</i> vs. <i>A. kitaibelii</i>	0.54	0.69	0.97	0.92	0.01	0.07	61.62	16.67
36	<i>Ablepharus budaki</i> vs. <i>A. pannonicus</i>	0.22	0.23	0.18	1	0.84	0.002	37.32	32.65
37	<i>Ablepharus budaki</i> vs. <i>A. rueppellii</i>	0.25	0.38	0.001	1	1	0.008	54.53	26.44
38	<i>Ablepharus budaki</i> vs. <i>A. sikimensis</i>	0.02	0.03	0.002	0.99	0.99	0.08	49.15	28.45
39	<i>Ablepharus chernovi</i> vs. <i>A. deserti</i>	0.36	0.40	0.88	0.92	0.11	0.09	50.04	30.55
40	<i>Ablepharus chernovi</i> vs. <i>A. grayanus</i>	0.28	0.31	0.68	0.99	0.34	0.01	57.08	18.27
41	<i>Ablepharus chernovi</i> vs. <i>A. himalayanus</i>	0.01	0.01	0.18	1	0.78	0.16	42.09	40.26
42	<i>Ablepharus chernovi</i> vs. <i>A. kitaibelii</i>	0.62	0.65	1	0.98	0.001	0.02	57.21	20.21
43	<i>Ablepharus chernovi</i> vs. <i>A. pannonicus</i>	0.39	0.52	0.62	1	0.42	0.005	56.42	18.44
44	<i>Ablepharus chernovi</i> vs. <i>A. rueppellii</i>	0.10	0.15	0.001	1	1	0.01	58.89	19.98
45	<i>Ablepharus chernovi</i> vs. <i>A. sikimensis</i>	0	0	0.96	0.82	1	1	48.69	35.37
46	<i>Ablepharus deserti</i> vs. <i>A. grayanus</i>	0.18	0.20	0.47	0.64	0.50	0.35	59.50	26.95
47	<i>Ablepharus deserti</i> vs. <i>A. himalayanus</i>	0	0	0.94	0.98	1	1	51.62	34.69
48	<i>Ablepharus deserti</i> vs. <i>A. kitaibelii</i>	0.14	0.16	0.001	0.58	1	0.42	57.21	20.21
49	<i>Ablepharus deserti</i> vs. <i>A. pannonicus</i>	0.32	0.42	0.19	1	1	0.41	52.12	31.74
50	<i>Ablepharus deserti</i> vs. <i>A. rueppellii</i>	0.009	0.01	0.001	0.97	1	0.07	61.03	20.12

Table 4 (continued)

No.	Species	Schoener's D	Hellinger's I	E (D)	S (D)	E (C)	S (C)	Correlation circle (%)	
								PC1	PC2
51	<i>Ablepharus deserti</i> vs. <i>A. sikimensis</i>	0	0	0.93	0.99	1	1	60.70	26.96
52	<i>Ablepharus grayanus</i> vs. <i>A. himalayanus</i>	0	0	0.95	1	0.03	0.79	60.48	30.70
53	<i>Ablepharus grayanus</i> vs. <i>A. pannonicus</i>	0.51	0.63	0.43	1	0.59	0.002	47.09	41.94
54	<i>Ablepharus grayanus</i> vs. <i>A. rueppellii</i>	0.06	0.14	0.14	1	0.80	0.003	49.74	20.52
55	<i>Ablepharus grayanus</i> vs. <i>A. sikimensis</i>	0	0	0.94	0.93	1	1	40.06	29.49
56	<i>Ablepharus himalayanus</i> vs. <i>A. kitaibelii</i>	0.08	0.08	0.77	0.98	0.28	0.03	60.09	26.45
57	<i>Ablepharus himalayanus</i> vs. <i>A. pannonicus</i>	0	0	0.92	1	1	1	59.70	20.97
58	<i>Ablepharus himalayanus</i> vs. <i>A. rueppellii</i>	0.04	0.08	0.05	1	0.94	0.01	51.21	36.01
59	<i>Ablepharus himalayanus</i> vs. <i>A. sikimensis</i>	0.10	0.21	0.009	1	1	0.003	65.93	22.64
60	<i>Ablepharus kitaibelii</i> vs. <i>A. pannonicus</i>	0.04	0.06	0.02	0.83	0.98	0.18	64.28	22.24
61	<i>Ablepharus kitaibelii</i> vs. <i>A. rueppellii</i>	0.28	0.40	1	1	0.001	0.002	51.91	30.01
62	<i>Ablepharus kitaibelii</i> vs. <i>A. sikimensis</i>	0.01	0.02	0.18	1	0.90	0.13	67.71	15.79
63	<i>Ablepharus pannonicus</i> vs. <i>A. rueppellii</i>	0.08	0.17	0.001	1	1	0.006	50.55	27.15
64	<i>Ablepharus pannonicus</i> vs. <i>A. sikimensis</i>	0	0	1	1	0.05	0.98	58.48	26.61
65	<i>Ablepharus rueppellii</i> vs. <i>A. sikimensis</i>	0.08	0.09	0.04	0.82	0.94	0.15	50.79	33.84

2006). However, when looking at climate change models, this distribution pattern is somewhat difficult to be neutral (Araújo et al., 2006). On the other hand, the population ecology of *Ablepharus* species has received less attention, and only a few studies have been published about their activities. Long-term recaptures of the *A. kitaibelii* by Vergilov et al. (2022) indicate that it is largely sedentary, perhaps not engaging in many moves over a distance of more than 60 m, and typically keeping a small range. Based on Table 3; Fig. 2 and Fig. S1, the present study, used the limitless dispersion hypothesis assumption, and forecasted range loss and gain for species by 2081–2100.

Based on ssp585, the eSDM findings show that 83.33% of the species (*A. alaiicus*, *A. anatolicus*, *A. budaki*, *A. chernovi*, *A. grayanus*, *A. himalayanus*, *A. kitaibelii*, *A. pannonicus*, *A. rueppellii*, *A. sikimensis*) contracted their recent distribution, particularly along their southern margins at lower altitudes. Meanwhile, they may also shown the northward expansion. On the other hand, some species expanded their range to north (*A. deserti*) or south (*A. bivittatus*). However, in the ssp126, most species' distribution ranges remain unchanged despite contractions and expansions (Fig. S1, Table 3). According to Moreno-Rueda et al. (2012), other factors, such as geographic barriers, may also have an impact on the rate of a species' migration to the north. Therefore, more investigation in these areas is needed for the species under consideration. Similar to these results, monitoring in the southern range of the species revealed that several lowland populations either became extinct in 10 years or saw a reduction in density of more than 50% during a warm spell (Le Galliard et al., 2012).

Climate-related species distribution models can help us predict how different species may react to future climate changes (Araújo et al., 2006). The results of this study showed that climatic variables affect the distribution of *Ablepharus* species through a wide distribution range. Seasonal temperature variations (for six species), precipitation during the year (for three species), maximum temperature of warmest month (for two species) and precipitation seasonality (for one species) are the primary factors contributing to the understanding of their distribution preferences (Table 2). This demonstrates how temperature and precipitation have a significant impact on where these species are found. Therefore, the results of this study may also shed new light on how future climate change may affect mid-latitude lizards. However, the genus *Ablepharus* has been influenced by a variety of biotic and abiotic factors that have shaped its distribution limits (Kurnaz, 2022). Therefore, it is important to understand that the models used in this study do not take into account additional factors that may affect lizard declines, such as habitat loss and fragmentation, anthropogenic pollution, parasitism and disease, and invasive species predation (Todd et al., 2010). For instance, several studies have shown that habitat loss and fragmentation have a detrimental effect on how lizard species disperse (Bell et al., 2010; Clark et al., 1999; Delaney et al., 2010; Stow et al., 2001; Templeton et al., 2001). Restricted migration can lead to inbreeding, decreasing populations, and a loss of genetic diversity (Andersen et al., 2004; Gilpin, 1991; Hastings & Harrison, 1994; Lacy & Lindenmayer, 1995; Leimu et al., 2010; Matesanz et al., 2017; Schlaepfer et al., 2018; Senior et al., 2021). Therefore, more study is necessary, especially in light of climate change, as there are no studies

on *Ablepharus* species in this area. Furthermore, non-climatic variables may play a significant role in forecasting taxonomic ranges, and their inclusion in models, along with feedback interactions between variables, is anticipated to enhance future estimations of species extinction or decline (Coudun et al., 2006; Luoto et al., 2006). Examples of such variables include habitat distribution in space, habitat management, nutritional considerations, and anthropogenic disturbance (Storfer, 2003). Therefore, a multi-factorial study would be required to explore these complex linkages.

If each species occupies its unique Grinnellian niche on a macro-scale, then ENMs approaches may be used to assess the potential distribution of a species derived from its realized niche (Elith & Leathwick, 2009; Guisan & Thuiller, 2005; Thrasher et al., 1917). ENMs may be used to evaluate speciation processes and even for species delimitation when combined with phylogenetic data (Hawltischek et al., 2011; Jakob et al., 2010). Because systematics has a significant biogeographical component, ENMs techniques are currently growing increasingly crucial in phylogenetic study (Raxworthy et al., 2007). The genus *Ablepharus*, however, received little attention from phylogenetic, paleogeographic, and phylogeographic investigations, and no one has performed a comprehensive analysis that incorporates ecological data. However, several examples of European organisms whose geographic range displays northern-southern shifts due to the response of the climatological oscillations (Villa & Delfino, 2019; Weiss & Ferrand, 2007) might be a suitable explanation for Mediterranean based *Ablepharus* species (*A. kitaibelii*, *A. ruepellii*, *A. budaki*, *A. chernovi*). On the other hand, it is believed that central Asia is the origin of the genus *Ablepharus* (Karamiani et al., 2021). As a result of it, Asiatic *Ablepharus* species diverged in the early Miocene and displayed various evolutionary history patterns (Karamiani et al., 2021). The dispersal routes of the genus are thought to be related to the geologic and tectonic formations of the Iranian Plateau, the rise of the Caucasus mountains, the collision of the Indian plate with Eurasia and the rise of the Himalaya mountains and the Tibetan plateau (Ghorbani, 2021; Tripathy-Lang et al., 2013). This situation enabled the distribution of the genus in a wide range, and as a result of it, local conditions, such as altitudinal preferences aided the speciation process of the *Ablepharus* species. Because the orogeny of the western Asia mountains might have provided different bioclimatological niche characteristics in lower and higher elevations. For instance, the warm and humid climate in northwest Iran caused a remarkable fragmentation of floristic and faunistic elements (Bennett, 1990) and as a consequence of it, most populations of *Ablepharus* had expanded or sheltered in cool habitats. The migration of *A. bivittatus* from northwest to lower latitudes

can be considered a case study for this fact (Karamiani et al., 2021).

A study on the populations of *A. kitaibelii* using two mitochondrial markers, cytochrome *b* and 16 S rRNA genes showed that *A. budaki* and *A. chernovi* taxa can be considered as valid species either subspecies as previously known (Poulakakis et al., 2005). In addition, another study by Poulakakis et al. (2013) examined the phylogenetic relationships between the *A. kitaibelii*, *A. budaki* and *A. chernovi* taxa through a limited and narrow distribution area including Cyprus, Syria and Türkiye countries. The *A. budaki*, *A. kitaibelii*, *A. chernovi*, *A. pannonicus*, and *A. ruepellii* species, which are regarded as the *A. kitaibelii* complex, were compared phylogenetically in subsequent research employing nuDNA, and mtDNA (Skourtanioti et al., 2016). This study suggested that the *A. anatolicus* taxon, which was a subspecies of the *A. budaki* taxon, is in fact a separate and valid species. Based on estimated divergence times, the complex originated in the Middle Miocene (around 16 Mya) and most species diverged in the Late Miocene. This hypothesis implies that the *A. kitaibelii* species complex originated in Anatolia by inferring that its current distribution pattern is a result of multiple dispersals and vicariance events associated with the Anatolian Diagonal, Mid-Aegean Trench, and mountain chain formation in southern and eastern Anatolia (Skourtanioti et al., 2016). Elsewhere, another recent phylogenetic and morphological study by Bozkurt and Olgun (2020) examined all *Ablepharus* taxa distributed in Türkiye and confirmed the full species status of *A. anatolicus*. These authors also suggested that *A. bivittatus* taxon should be included in the genus *Asymblepharus*. On the other hand, Karamiani et al. (2021) investigated the phylogeny of *Ablepharus* species from Iraqi and Iranian populations and mentioned the existence of a phylogenetically distinct species found in the eastern Iraq regions and the western and southwestern Iran regions. Hence, Mirza et al. (2022) while introducing the new genus *Protoablepharus*, revealed the necessity of considering all species belonging to the genus *Asymblepharus* as a synonym of *Ablepharus*. Our study clarifies the significance of niche conservatism and divergence in shaping diversity patterns and provides insight into the evolutionary history of the genus *Ablepharus*. We recommend that future investigations that take into account microhabitat and other parameters may be necessary to fully comprehend these speciation processes. They might also allow for new phylogenetic analysis studies on evolution and climate.

The development of climatic niche divergence occurs when the environmental conditions (such as elevation variations) are mutually unfavourable for adjacent populations. This can limit gene flow between the populations, leading to reproductive isolation and then speciation (Jezkova &

Wiens, 2018). Populations distributed parapatrically along a climatic gradient are the most evident geographic mode of speciation through climatic-niche divergence. In climatic-niche conservatism, different populations of a species are separated by a zone of unfavourable climatic conditions, either via dispersal over this zone or vicariance (i.e., populations that formerly occurred within the zone of now unsuitable conditions go locally extinct) (Hua & Wiens, 2013). Niche conservatism (climatic or not) may be the main force behind allopatry and, consequently, the beginning of allopatric speciation. Understanding the degree to which closely related and/or partially sympatric taxa deviate in their niche characteristics is necessary for identifying the processes driving broad-scale biogeographic patterns (Cuervo et al., 2021; Wiens et al., 2010). Our findings supported the idea that ecology played a role in speciation by demonstrating that closely related *Ablepharus* taxa occur in and respond to climatic features of various habitats (Zink, 2014). According to the findings of this study, both niche divergence and niche conservatism can contribute to the evolution of these species. Niche equivalency and similarity test based on Schoener's D statistic and Hellinger's I in E-space show that the actual Grinnellian niches of the 28 paired comparisons evaluated in this study had significantly divergent potential distributions (see Table 4). There is a possibility that the current pattern of distribution is the result of vicariance-driven allopatric speciation followed by secondary sympatry. As a result of the niche overlap test, the niche equivalency and similarity tests, which involved 11 and 21 pairwise comparisons, niche conservatism is supported by the fact that the biological heterogeneity between these species results from an allopatric speciation process (see Table 4). This strongly implies that biotic interactions have an impact on the process of niche diversification.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s11692-023-09603-6>.

Author Contributions All authors contributed to the study conception and design. Material preparation and data collection were performed by Somaye Vaissi, Muammer Kurnaz, Mehmet Kürşat Şahin, analysis was performed by Somaye Vaissi. The first draft of the manuscript was written by Somaye Vaissi, Muammer Kurnaz, Mehmet Kürşat Şahin and Axel Hernandez and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

Data Availability All data generated or analyzed during this study are included in this published article and its supplementary information files.

Declarations

Conflict of interest Its preparation was not supported by any external funding source, nor are there any conflicts of interest in connection with our authorship.

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